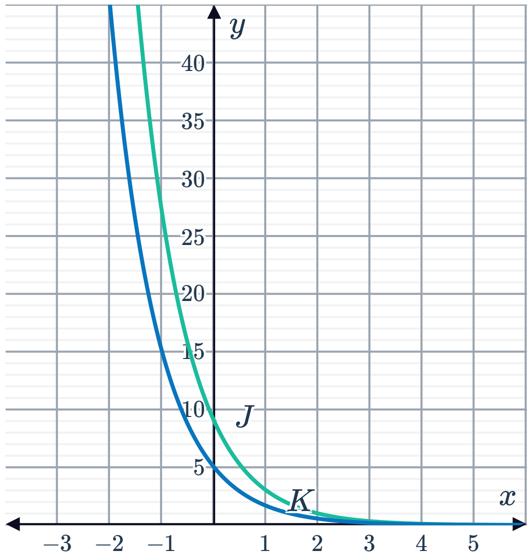
Comparison of exponential functions

1

a Decreasing

b



c As $x \to \infty$, functions $J, K \to 0$.

d Dilate $y = 3^{-x}$ vertically by a factor of 5.

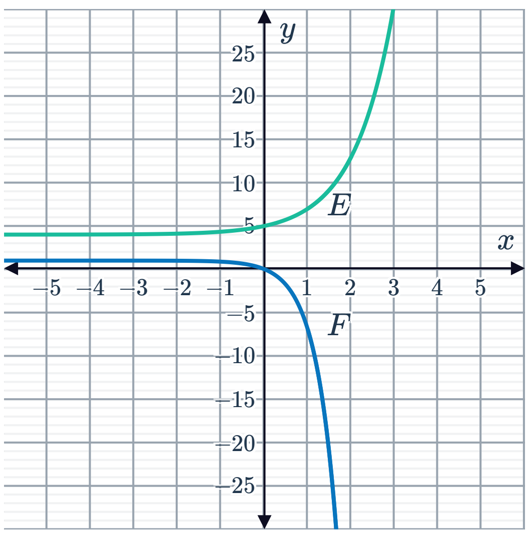
2

a Once

b Function P

3

a



b Zero times

c As $x \to \infty$, the function $E \to \infty$ and function $F \to -\infty$.

Applications

4

a 8 million

b
i 7130 million
ii 9286 million

5

a
i $A = P(1.053)^t$
ii $A = P(1.0125)^{4t}$

b 5.09%

c Bank A

6

a Sale price of property

b 5.3%

c -5.3%

7

a $M = 60\,000\,(1.032)^t$

b $P = 50\,000\,(1.061)^t$

c Mohamad's salary will be $\$1153.51$ greater than Elizabeth's.

d Elizabeth's salary

8

a 30

b 40%

c 100%

d The population of bacteria in the dark environment is increasing at a faster rate than the population of bacteria in the sunlit environment.

9

a $P = 90 + 5t$

b $Q = 90 \times (1.04)^t$

c The current company

d Opening his own business

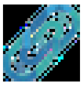
10

a $P = 100\,(0.85)^t$

b Fabian

c No, neither Fabian or Zainab will completely forget the content. There is a horizontal asymptote at $y = 0$ and so the recall percentage for both will never reach zero.

11 a Yes, there will be a food shortage.



We can derive the equation $F = 3\,374\,000\,(1.02)^n$ to represent the growth of the population, and the equation $P = 2\,410\,000\,(1.022)^n$ to represent the number of people the food supply will feed.

We can see that the food production is increasing, but at a slower rate than the population's rate of increase, so the population will eventually exceed the amount that can be fed.

b We can draw the graph of both functions, using technology to find any points of intersection. If the y-value of the population is greater than the y-value of the population that can be fed for the same value of n then there will be a food shortage.

12

a 179

b 132

c Blair

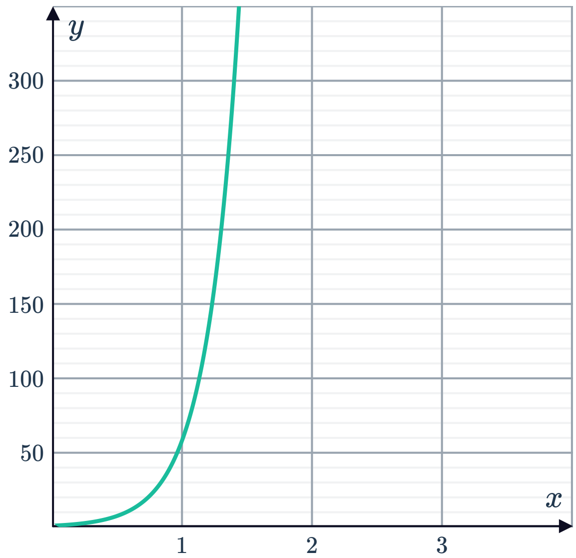
d 70%

e If you have a high sugar diet, your blood sugar level peaks at greater amounts and decreases more rapidly than a low sugar diet.

13

a The population of bacteria J increases faster than the population of K.

b



C